

Fine-Tuning Silently Breaks Linear Safety Monitors

Varun Iyer

Abstract

Safety researchers propose monitoring AI models by training linear probes on internal activations, then deploying those probes as runtime monitors. This assumes the probe direction remains valid as the model is trained further. We test that assumption by tracking a shortcut-detection probe across five rounds of supervised fine-tuning on Qwen2.5-Coder-7B. The probe direction rotates 40–70° within a single training step, yet a fresh probe at the same checkpoint recovers full accuracy—the feature has not disappeared, only moved. The standard validation protocol misses this: additive steering along the old direction still produces clean dose-response curves, reporting that the direction “still works.” But the Arditi et al. [3] necessity test—projecting the direction out of the residual stream—produces no behavioral change. The direction pushes behavior around via magnitude without being the axis the model uses. This is a compound failure invisible to any single diagnostic. Only the ablation test catches it, and only difference-of-means extraction (not logistic regression) retains causal necessity through training.

1 Introduction

A growing body of work proposes deploying linear probes as safety monitors: train a classifier to detect a concerning behavior from a model’s internal activations, then run it at inference time to flag or steer away from that behavior [1, 3, 10, 12, 13]. The pitch is compelling—probes are cheap, interpretable, and effective at the checkpoint where they are trained. But every deployment of a probe-based monitor implicitly bets that the direction it found is a property of the feature, not a coincidence of the checkpoint. None of the papers that ship these tools test that bet under training pressure. We do.

Representation Engineering [13] extracts concept directions at a single checkpoint. ActAdd [12] deploys steering vectors extracted and applied at the same checkpoint. The refusal-direction result [3] pairs additive steering with directional ablation, but on a single frozen model. The Mythos system card [2] reports the phenomenon as an operational difficulty—“the causal effects of individual features often changed over the course of post-training”—without measuring the rotation angle or the sufficiency–necessity dissociation. This paper provides the controlled measurement: same probe, same trajectory, three interventions.

Setting. We train Qwen2.5-Coder-7B [9] with rank-16 LoRA [8] on CodeContests problems [11], rewarding solutions that pass the *visible* test cases whether or not they pass the hidden ones. Solutions that pass visible but fail hidden tests are “shortcuts” [4]; hidden tests provide an objective, label-free ground truth. We iterate for five SFT rounds and track a linear shortcut/general probe at layer 11 across all six checkpoints (base plus five rounds).

Contributions.

1. **Direction drift within one SFT step (§4).** Across three linear extraction methods, the cosine between the base direction and every post-base direction is substantially below 1.0. Under

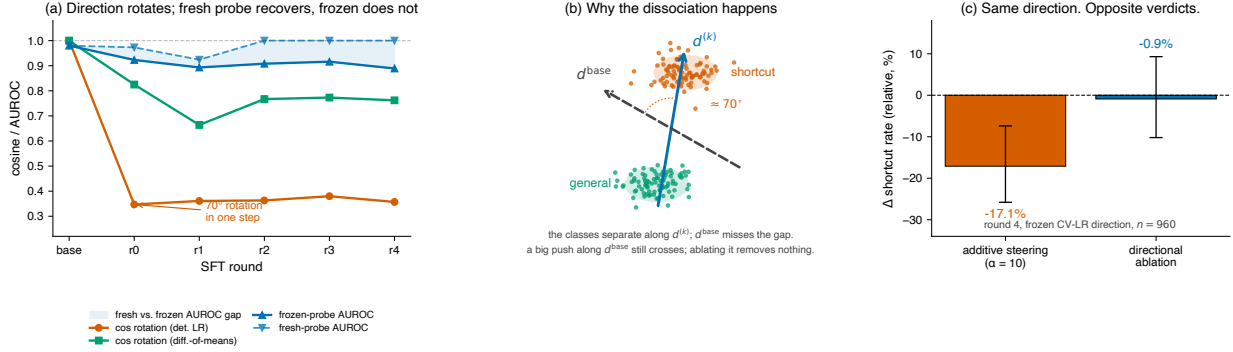


Figure 1: **A frozen probe’s coordinates move within one SFT step, but additive steering still reports success.** (a) Under deterministic LR, $\cos(d^{base}, d^{(k)})$ collapses to ≈ 0.35 at round 0 and stays there; under difference-of-means the rotation is smaller ($\approx 40^\circ$) but present. The shaded region is the gap between the fresh probe (refit at each checkpoint, recovers AUROC ≥ 0.92) and the frozen probe (fit at base, degrades to 0.89). The feature persists; the coordinates move. (b) Geometric intuition: the shortcut and general clusters retain their linear separation, but the extracted probe vector rotates off the class-separation axis. A large additive push along d^{base} still crosses the new boundary by virtue of magnitude; projecting d^{base} out of the residual stream removes almost nothing causal. (c) The dissociation on the same direction at the same checkpoint: additive $\alpha=10$ gives -17.1% , ablation gives -0.9% . Error bars are bootstrap 95% CIs, $n=960$.

deterministic LR, $\cos(d^{base}, d^{(k)}) \approx 0.35$ for every $k \geq 1$ (a $\approx 70^\circ$ rotation that arrives in one step and stays put). Under difference-of-means, $\cos \in [0.66, 0.82]$ ($\approx 40^\circ$). Fresh probes recover AUROC ≥ 0.92 on the rotated subspace; frozen AUROC drops to 0.89.

- Additive steering survives the rotation (§5).** Frozen-direction steering at round 2 and round 4 produces monotonic dose-response reductions in shortcut rate. Round 4, $\alpha=10$: -17.1% , 95% CI $[-25.8, -7.4]$. Read alone, this is a clean “steering still works” result.
- Directional ablation dissociates from additive steering (§6).** Projecting the same frozen CV-LR direction out at round 4 produces -0.9% ($p=0.57$). The deterministic-LR frozen direction also nulls ($+2.4\%$). Only difference-of-means retains measurable necessity (-7.2% frozen; -36.5% fresh, the largest effect in the experiment). **Additive steering alone does not license necessity claims.**

2 Related work

Linear probing and activation steering. Linear probing [1, 7] reads concepts from hidden activations via linear classifiers. Representation Engineering [13] extracts concept directions at a single checkpoint using LAT (PCA on contrastive differences), difference-of-means, or supervised probes. ActAdd [12] extracts steering vectors from contrastive prompt pairs and adds them at inference. Ardit et al. [3] show that refusal is mediated by a single direction, extracted via difference-of-means and validated by *both* additive steering and directional ablation. All three operate at a single checkpoint. A notable commonality: RepE and Ardit both default to difference-of-means (or LAT reducing to it) rather than a trained LR probe, on stability grounds. Our §4 confirms why: CV-selected LR weight vectors land at very different points along the regularization path

depending on fold tie-breaks, producing stability metrics that reflect the CV’s behavior as much as the feature’s.

Feature stability under optimization. The Mythos system card [2] documents SAE features for concealment, strategic manipulation, and transgressive action that persist during alignment training but with shifted causal effects on behavior (“the causal effects of individual features often changed over the course of post-training, making it difficult to attribute behavioral changes merely to increases or decreases in particular feature activations”). Mythos reports the phenomenon qualitatively on SAE features in a production alignment run; it does not measure the rotation angle, the fresh-vs-frozen recovery gap, or the additive-ablation dissociation. Our contribution is the controlled analog: probe-derived directions, checkpointed trajectories, and the three-way intervention battery on the same direction at the same checkpoint.

AI control and monitor robustness. Greenblatt et al. [5] define AI control as designing deployment protocols that remain safe even if the target model is intentionally subversive. Our setting is explicitly non-adversarial; the direction drift we observe arrives as an incidental side-effect of task-reward optimization, which is an easier-than-worst-case scenario for the monitor. The adversarial version—gradient pressure against a frozen direction—is the natural follow-up (§8).

Shortcut learning. Shortcut learning [4] in code-generation models [6] manifests as solutions that pattern-match visible test cases without implementing the general problem. We use CodeContests because hidden tests provide objective ground truth without human labels.

3 Setup

Model and training. Qwen2.5-Coder-7B-Instruct in bfloat16. LoRA adapters at rank 16, $\alpha=16$, targeting all projection matrices. Iterative rejection-sampling SFT: generate $N=8$ completions per problem with vLLM, filter to those that pass all public tests, SFT on the filtered pool (pooled across all prior rounds) for one epoch at learning rate 2×10^{-5} , repeat for five rounds.

Dataset. CodeContests [11], filtered to Codeforces ratings 800–1800 with ≥ 3 public tests and ≥ 10 hidden tests, excluding interactive problems and lenient checkers. A 1000-problem subset is frozen with SHA-256 fingerprint and used for every run. Shortcut rate on the base model is 4.5% per generation across 16,000 base-model generations; 242 of 1000 problems produce at least one shortcut and at least one general solution.

Probe. A linear probe is trained via `LogisticRegressionCV` with 5-fold `GroupKFold` grouped by problem ID, on layer-11 residual-stream activations (mean-pooled over completion tokens) with balanced shortcut/general classes. We save the unscaled weight vector as the probe’s direction. For method-sensitivity checks (§4) we also extract deterministic LR at fixed $C=1$ and difference-of-means between the balanced class means.

Steering. A forward hook on the layer-11 output subtracts $\alpha \cdot \hat{d}$ from the residual stream at every token during generation. We evaluate at $\alpha \in \{2, 5, 7, 10\}$ on 30 shortcut-producing problems with 32 completions each ($n=960$ per condition).

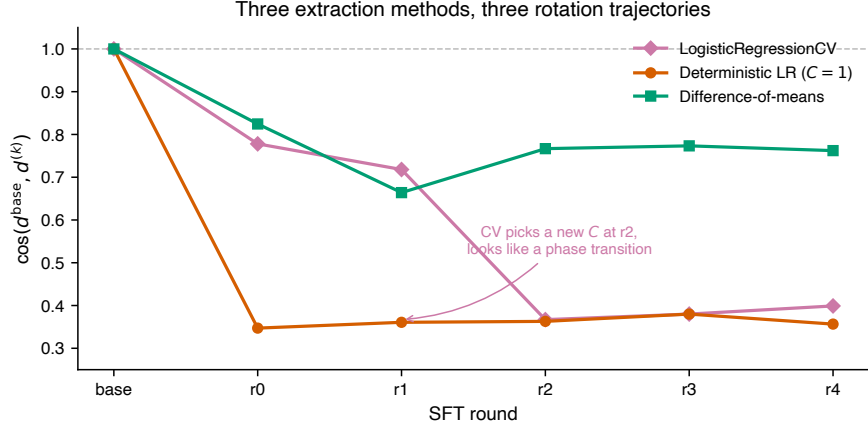


Figure 2: **Three extraction methods, three rotation trajectories, one conclusion.** All three agree the direction has moved by round 4; they disagree on the shape. The gradual CV-LR decline is an artifact of `LogisticRegressionCV` selecting C values six orders of magnitude apart as fold-level tie-breaks shift across checkpoints (§4). Under a fixed C , the rotation collapses to a single jump at round 0.

Ablation. Following Arditi et al. [3], we replace $h \mapsto h - (h \cdot \hat{d})\hat{d}$ at every token and every forward pass at layer 11, then generate with the resulting intervened model. All other conditions match the steering protocol.

Probe validation. Leave-one-problem-out cross-validation of the base-model probe: mean AUROC 0.924, median 0.975, min 0.328, std 0.110. The minimum-AUROC fold is a structurally anomalous problem (closed-form shortcut, iterative general solution; the modal CC pattern is the opposite), which we return to in §8. The confidence-direction alignment with the probe is 0.027, ruling out the probe detecting model confidence.

Methodological note. GroupKFold can mask severe problem-identity leakage when group counts are small: a probe can reach perfect in-sample AUROC while catastrophically failing on held-out groups. We recommend LOO validation as a standard check whenever group counts fall below ~ 30 .

4 The readable direction rotates within one SFT step

Under every extraction method we tested, the direction is no longer where it was after one SFT round. Fresh probes at each checkpoint recover $\text{AUROC} \geq 0.92$ —the feature has not disappeared; only its coordinates have moved. The feature does not decay: it becomes *more* separable, not less.

Table 1 reports the core measurements. Across three extraction methods on identical activations, the direction has rotated by round 4. The shape differs. Under `LogisticRegressionCV`, the cosine shows a gradual decline $1.00 \rightarrow 0.78 \rightarrow 0.72 \rightarrow 0.37 \rightarrow 0.38 \rightarrow 0.40$ with an apparent “phase transition” between round 1 and round 2. Under deterministic LR at fixed $C=1$, the cosine drops to 0.35 at round 0 and stays at 0.35–0.40 for every subsequent checkpoint. Under difference-of-means, the cosine drops to 0.82 at round 0 and fluctuates in $[0.66, 0.77]$ thereafter.

method	base	r0	r1	r2	r3	r4
LogisticRegressionCV	1.00	0.78	0.72	0.37	0.38	0.40
Deterministic LR ($C=1$)	1.00	0.35	0.36	0.36	0.38	0.36
Difference-of-means	1.00	0.82	0.66	0.77	0.77	0.76
Frozen probe AUROC (LR)	0.980	0.923	0.893	0.908	0.916	0.889
Fresh probe AUROC	0.980	0.973	0.924	1.000	1.000	1.000

Table 1: $\cos(d^{\text{base}}, d^{(k)})$ and probe AUROC across six checkpoints. **Fresh probes recover AUROC ≥ 0.92 at every checkpoint**—the feature is still linearly separable, in a different direction. LOO cross-validation of the fresh probe at round 4: mean AUROC 0.955, median 1.000. Shortcut-strategy distribution (iterative vs. non-iterative) shifts by only 4.4 pp across rounds (permutation $p \approx 0.57$), ruling out behavioral drift as the cause of the frozen AUROC decline.

Why the trajectories diverge. Cross-method cosines at step 0 on identical activations show that the three methods already point in three different directions at base: CV-LR vs. deterministic LR is $\cos=0.65$; CV-LR vs. DoM is $\cos=0.53$. `LogisticRegressionCV` picks C via fold-level tie-breaks, which land at values six orders of magnitude apart as the near-separability of the fold data changes across checkpoints (step 0: $C \approx 21.5$; step 2: $C \approx 7.7 \times 10^{-4}$; step 3: $C \approx 21.5$ again). The C re-selection moves the direction along the regularization path independently of any change in the feature. Under a fixed C , this degree of freedom is closed off and the rotation collapses to a single jump at round 0.

A frozen monitor deployed at base and never refit would silently lose calibration within one SFT step.

Ruling out behavioral drift. An alternative explanation for the frozen AUROC drop is that shortcut strategies themselves change over training. We classify each shortcut as iterative (contains loops) or non-iterative (straight-line/closed-form). The iterative fraction ranges from 66.0% to 70.4% across the six checkpoints (range 4.4 pp, permutation test $p \approx 0.57$): a random walk within sampling noise. An upper-bound calculation puts strategy-drift’s contribution to the AUROC drop at ≤ 3 pp; the observed drop is 9 pp. Representational drift, not behavioral drift, is the dominant contributor.

5 Additive steering survives the rotation

We ran the full α -sweep at round 2 (the apparent peak of a preliminary trajectory) and round 4 (the final checkpoint). Both sweeps produce monotonic dose-response curves with significant effects at $\alpha \geq 5$.

Read alone, Table 2 says the frozen direction is a reliable causal handle: monotonic dose-response at both checkpoints, significant effects at $\alpha \geq 5$, strongest single-condition effect at round 2, $\alpha=10$ (-24.5% , $p(\Delta < 0) > 0.999$). The round 4 curve is quantitatively less responsive at every dose but still reliably negative at $\alpha=10$.

A single noisy measurement almost anchored a wrong narrative. The initial round 4, $\alpha=5$ measurement returned $+0.7\%$ (95% CI $[-9.2, +11.8]$), conspicuously off the monotonic curve. An independent re-measurement of the identical condition returned -9.8% (CI $[-18.4, -0.2]$). The two runs disagreed by the full binomial CI width; the re-measurement sits on the monotonic curve,

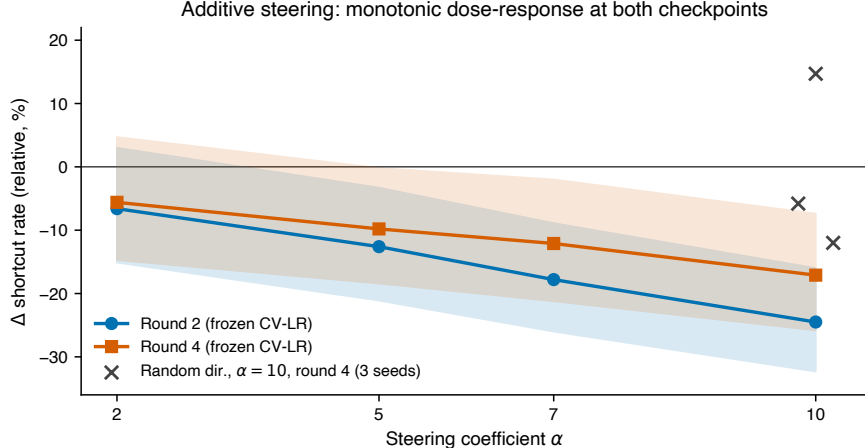


Figure 3: **Frozen-direction additive steering is well-behaved at both checkpoints.** Monotonic dose-response in relative shortcut rate across $\alpha \in \{2, 5, 7, 10\}$ at round 2 and round 4. Shaded bands are bootstrap 95% CIs, $n=960$ per condition. Grey crosses: three independent norm-matched random directions at $\alpha=10$, round 4. Additive steering alone is consistent with both direction specificity and magnitude; only the ablation test in §6 discriminates.

α	Round 2 (Δ rel, 95% CI)	Round 4 (Δ rel, 95% CI)
2	-6.6% [-15.1, +3.0]	-5.6% [-14.7, +4.7]
5	-12.6% [-21.1, -3.3]	-9.8% [†] [-18.4, -0.2]
7	-17.8% [-26.0, -8.9]	-12.1% [-21.2, -2.0]
10	- 24.5% [-32.3, -16.0]	-17.1% [-25.8, -7.4]

Table 2: Additive-steering dose-response with the frozen CV-LR direction at round 2 and round 4. Each row is an independent $n=960$ run with its own unsteered baseline. [†]The round 4, $\alpha=5$ value is a re-measurement (see below).

the original was at its edge. At $n \approx 10^3$ with single-digit effects, one hour of A100 compute was the difference between publishing a “dead zone” narrative and not.

Norm-matched random control. To test whether any large perturbation at matched norm would suppress shortcuts, we ran three Gaussian random directions scaled to the frozen direction’s L2 norm (1.51), at $\alpha=10$, round 4: seed 41 -5.8%, seed 42 +14.7%, seed 43 -12.0%. Mean across seeds: -1.0%. The frozen direction’s -17.1% sits outside this 3-seed envelope on the low end by 5 pp, but the envelope is wide. A more decisive discrimination comes from replacing *the intervention* rather than the direction: §6.

6 Directional ablation dissociates from additive steering

Additive steering tests *sufficiency*: a large-enough component along the direction changes behavior. Directional ablation— $h \mapsto h - (h \cdot \hat{d})\hat{d}$ at every layer-11 token—tests *necessity*: if we remove the component along $\hat{d}$ entirely, does behavior change? The two tests come apart on the same direction at the same checkpoint (Table 3).

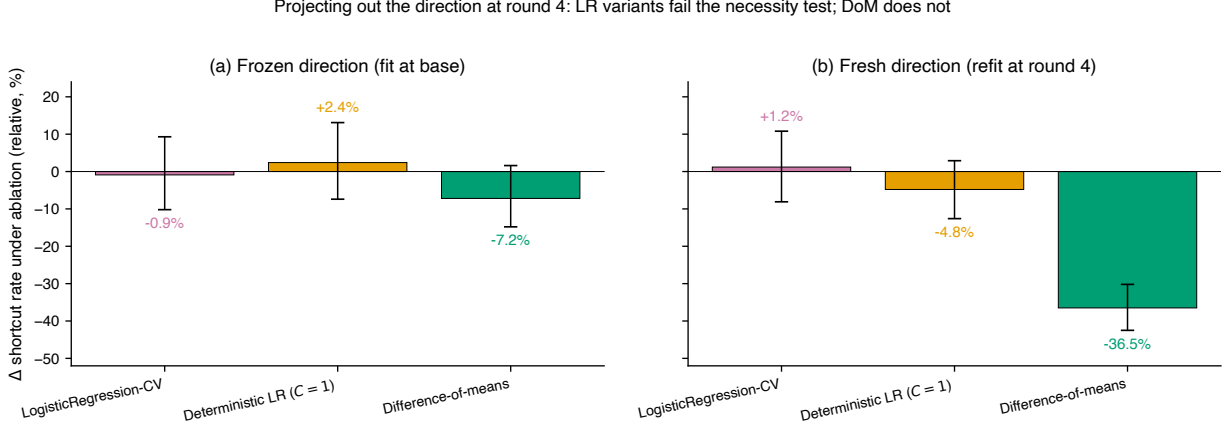


Figure 4: **Projecting the direction out of the residual stream at round 4.** (a) Frozen directions (fit at base). Both LR variants fail the necessity test; only difference-of-means shows measurable effect. (b) Fresh directions (fit at round 4). LR variants again fail or are null; DoM produces -36.5% , the largest single-condition effect in the experiment. $n=960$ per condition; bootstrap 95% CIs. The Arditi necessity test’s sign and magnitude depend strongly on extraction method, and for LR-family directions disagree with the verdict that additive steering (Fig. 3) returns.

variant	method	fit at	Δ abs	Δ rel	$p(\Delta < 0)$
frozen	CV-LR	base	-0.4 pp	-0.9%	0.57
frozen	det. LR	base	$+1.0$ pp	$+2.4\%$	0.31
frozen	DoM	base	-3.1 pp	-7.2%	0.92
fresh	CV-LR	r4	$+0.5$ pp	$+1.2\%$	0.39
fresh	det. LR	r4	-2.2 pp	-4.8%	0.83
fresh	DoM	r4	-15.9 pp	-36.5%	1.000

Table 3: Directional ablation at round 4 across extraction methods. The CV-LR frozen direction—the same direction whose additive $\alpha=10$ effect is -17.1% (Table 2)—shows -0.9% under ablation. The fresh DoM direction shows -36.5% , the largest single-condition effect in the paper.

Five observations.

(1) **Both CV-LR directions fail the Arditi necessity test—frozen and fresh.** Projecting the CV-LR frozen direction out at round 4 produces $\Delta = -0.9\%$ ($p=0.565$). Projecting the *fresh* CV-LR direction—the one with in-sample AUROC 1.000—produces $\Delta = +1.2\%$. A probe with AUROC 1.0 can be completely causally disconnected from the feature it classifies.

(2) **Fixed- C LR shows an intermediate pattern.** Frozen $+2.4\%$, fresh -4.8% . Freshness helps; removing CV’s C -selection instability helps; neither closes the gap to DoM.

(3) **Difference-of-means is the only extraction method that produces decisive causal necessity at round 4.** The frozen DoM direction, fit on step 0 data and never refit, still produces -7.2% ($p=0.915$). The fresh DoM direction produces -36.5% ($p=1.000$): larger than any additive effect at any α in §5, including the -24.5% headline at round 2, $\alpha=10$. The feature has not decayed; it has reorganized into a sharper, more causally concentrated subspace.

(4) The additive-ablation dissociation on the CV-LR frozen direction is the load-bearing finding. Additive steering gives -17.1% with a CI that excludes zero. Directional ablation on the same direction at the same checkpoint gives -0.9% with 43% of its CI mass above zero. Sufficiency and necessity return opposite verdicts. The Arditì necessity test is precisely the one single-direction claims are meant to survive, and it is the one this LR-family direction fails.

(5) The direction additive steering is operating on is not the direction the model uses. Intuitively, a logistic regression classifier in 3584 dimensions finds the hyperplane that best separates the classes by combining thousands of tiny signals across all dimensions. The class-mean difference captures a single dominant axis. In high dimensions these are nearly orthogonal—the classifier spreads its weight across thousands of small signals, most of which are orthogonal to the single dominant axis of class separation—so on step 0 activations, $\cos(\text{LR}, \text{DoM}) \approx 0.26$ with the standard scaler and 0.37 without—so projecting out the LR direction removes almost none of the mean-shift signal that the model actually uses. An additive push at $\alpha = 10 \cdot \|\hat{d}\|$ crosses the decision boundary by virtue of magnitude; the decision boundary is not at the direction the push points along. The ablation test is what discriminates this from direction specificity.

Is this a scaler artifact? A plausible explanation for LR’s ablation failure is that the `StandardScaler` preprocessing upweights low-variance dimensions that downstream layers do not read from. We ran LR ablation with and without `StandardScaler` at both step 0 (frozen) and round 4 (fresh). Removing the scaler nudged the frozen direction from $+2.4\%$ to -2.7% , but nudged the fresh direction from -4.8% to $+1.4\%$: no consistent improvement. What is consistent is that *no LR variant across six conditions* approaches DoM’s ablation effect (best LR: -4.8% ; worst DoM: -7.2%). The gap is robust to preprocessing; the mechanism is the max-margin-vs.-mean-shift divergence of LR and DoM in high dimensions.

What this means for the sufficiency claim. The additive steering result of §5 remains factually true: at $\alpha \geq 5$, the frozen CV-LR direction produces monotonic reductions in shortcut rate. What the ablation test reveals is that the reduction operates via magnitude (a high-norm push along a correlated direction crosses the boundary) rather than direction specificity. The same “we can steer behavior” observation is consistent with the direction being the causal axis *or* being an innocuous-looking proxy; only the ablation test separates them.

7 Discussion

Additive steering alone does not license necessity claims. A monotonic dose-response curve is a sufficiency result, consistent with the steered direction being the causal axis *or* being a correlated proxy whose large- α push happens to cross a decision boundary. Our round 4 CV-LR comparison is a worked example: the same direction that produces a -17.1% dose-response reduction under additive steering produces -0.9% under ablation. Any deployment that relies on a steering vector as a controller should run the projective-ablation test alongside it before trusting that the vector is the handle it appears to be.

The compound failure the standard protocol misses. A frozen probe’s classification accuracy degrades from 0.98 to 0.89 (noticeable but not alarming); additive steering along the frozen direction still produces monotonic dose-response curves (reassuring); but the direction has lost causal necessity (invisible without an ablation test). A deployment team running the standard

evaluation protocol—probe AUROC plus steering validation—would see a slightly degraded but functional monitor. Only the ablation test reveals that the direction is no longer the handle it appears to be. This is the failure mode a monitor-based control protocol needs to rule out. The tools shipped by the current literature—probing and additive steering—are not individually equipped to detect it.

Extraction method is a first-order methodological choice. The rotation magnitude differs by a factor of ~ 2 between deterministic LR ($\approx 70^\circ$) and DoM ($\approx 40^\circ$), and the ablation verdict on the same frozen direction differs by a factor of ~ 8 (-0.9% for CV-LR, -7.2% for DoM). The extraction method that RepE and Arditì already default to on stability grounds is also the one that retains causal faithfulness at round 4. Single-direction claims should be reported for DoM (or LAT reducing to it) and, if desired, a regression-based direction *in addition*—not as a substitute. LR-direction AUROCs above 0.9 can coexist with total loss of projective-ablation effect.

The single-point noise cautionary tale. At $n=960$, the binomial CI on a single Δ is ± 3 pp—wider than many single-digit-percentage-point effects in this literature. Our round 4, $\alpha=5$ near-miss (two independent runs returning $+0.7\%$ and -9.8%) is a concrete demonstration that a single-point measurement at the edge of its CI can support a qualitatively different interpretation from the true effect. Before treating a surprising single-point comparison as a finding: re-run it, or sweep around it.

8 Limitations and the adversarial follow-up

Single feature, single model, single training recipe. One probe-derived feature (shortcut/general) in one model (Qwen2.5-Coder-7B), one training method (iterative rejection-sampling SFT), one task (CodeContests). Generalization to refusal, reasoning-step markers, emotion vectors, and to PPO or constitutional AI is not established. This is an existence proof for the failure mode at one point in the setting space.

Capability-entangled feature. “Shortcut” vs. “general” code is structurally entangled with “buggy” vs. “correct” code, a central capability of a code model. A feature more cleanly orthogonal to core capability (refusal, emotion, evaluation awareness) may behave differently under the same training pressure. This is the most important axis on which the finding needs replication.

Narrow probe target. The probe passes LOO at mean AUROC 0.924 but the per-problem breakdown reveals it captures the modal CC shortcut texture (iterative brute-force), not abstract shortcut-awareness. The one problem where it fails (AUROC 0.328) has the shortcut/general roles swapped relative to the modal pattern.

Sample size on individual comparisons. At $n=960$, the binomial 95% CI on a single Δ is ± 3 pp. Fresh-vs-frozen A/B comparisons we measured have CIs that cross zero and are reported descriptively. The main-text findings (rotation, dose-response, dissociation) rest on comparisons where CIs exclude zero or the effect is large enough that sample-size uncertainty is subordinate.

Ablation measured at a single checkpoint. We ran directional ablation at round 4 only. Whether the dissociation is present at earlier checkpoints (including round 2, which is more responsive to additive steering) is untested here.

Linear methods only. Non-linear probes (MLPs) and non-linear steering may behave differently. Our linear setup is a cleaner test of the linear-direction deployment assumption but does not bound what non-linear methods can do.

The adversarial follow-up. Our setting is non-adversarial: no gradient pressure against the monitor during training. The natural next experiment adds exactly that: (i) fit a frozen shortcut probe at base; (ii) during SFT, add a term to the loss that rewards lowering the probe’s score on shortcut completions; (iii) at each checkpoint, measure both the frozen and a fresh probe’s AUROC, and run the full additive+ablation battery. The prediction falls out of our result: a frozen probe subjected to direct gradient pressure against its readout will lose AUROC faster than under non-adversarial SFT, and the additive–ablation dissociation will appear earlier and larger. Validating that prediction would convert the present existence proof into a quantitative monitor-robustness benchmark relevant to AI control.

9 Conclusion

The cheapest methodological change that would catch this compound failure in practice: refit the probe whenever retraining occurs, and run the projective-ablation test alongside any steering-based validation before trusting the direction. When directions must be extracted, difference-of-means retains causal necessity where logistic regression—even at AUROC 1.0—does not.

The interpretability-for-safety agenda depends on directions being stable properties of features. Our result shows they are not, under the mildest form of training pressure—non-adversarial, single-step SFT. Whether adversarial training pressure accelerates the failure is the natural and most important follow-up.

Reproducibility. All experiments use a SHA-256–fingerprinted CodeContests subset, bfloat16 training and inference, and a fixed evaluation harness. Code and directional artifacts are available at the project repository.

References

- [1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.
- [2] Anthropic. Claude mythos preview system card. 2026. Section 4.5: White-box analyses of model internals.
- [3] Andy Arditi, Oscar Obeso, Aaquib Sylejmani, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*, 2024.
- [4] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2020.
- [5] Ryan Greenblatt, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. Ai control: Improving safety despite intentional subversion. *arXiv preprint arXiv:2312.06942*, 2023.

- [6] Dan Hendrycks, Steven Basart, Saurav Kadavath, Mantas Mazeika, Akul Arora, Ethan Guo, Collin Burns, Samir Puranik, Horace He, Dawn Song, and Jacob Steinhardt. Measuring coding challenge competence with apps. *NeurIPS Datasets and Benchmarks*, 2021.
- [7] John Hewitt and Christopher D. Manning. A structural probe for finding syntax in word representations. In *NAACL*, 2019.
- [8] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuezhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- [9] Binyuan Hui, Jian Yang, Zeyu Cui, et al. Qwen2.5-coder technical report. *arXiv preprint arXiv:2409.12186*, 2024.
- [10] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *NeurIPS*, 2023.
- [11] Yujia Li, David Choi, Junyoung Chung, et al. Competition-level code generation with alpha-code. *Science*, 378(6624):1092–1097, 2022.
- [12] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.
- [13] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023.